Phase Wave Cosmology: The Complete Theory

One medium, one yield, one formula, one ledger — from the femtometer knot to the Navier–Stokes cap

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Abstract

Space is not a zero. It is a continuous, mass-bearing electromagnetic medium: a birefringent lattice of paired waves held apart by heat, with a finite speed limit c_0, a finite density ceiling $\rho_{\max}$, and a finite yield Y_{bow}. Matter is this same medium tied into 720° toroidal knots, $m = L/c_0^2$, where L is the *light energy* [1]. Light is an unpaired wave. Black holes are shells at the density ceiling, terminating at a complete stop — not a singularity. Every constant of the theory is derived from the medium's locked set $\{\rho_0, \rho_{\max}, c_0, \hbar\}$ or declared as a calibration with its test record; the thermodynamic ledger closes to 1.000; the kinematic ladder $C(\rho) = \min(c_0, \sqrt{2Y_{\text{bow}}/\rho})$ admits zero violations among every catalogued object; and the gravitational coupling emerges as $g = 20^{1/16} = 1.205909$, within 0.069% of the charge-geometry value $16\pi^3\alpha/3$ [5, 6]. The theory is falsifiable, and its falsifiers are stated with every prediction.

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1 The medium and its laws

1.1 Ontology

- The universe is **one substance**: an electromagnetic field carrying mass, at different spacings. "Always been a solid; it was just how far apart it was." There are no separate materials, no vacuum, no zeros.
- The resting medium is **paired EM waves** held apart by heat. The speed of the medium's own waves is $c_0 = 1/\sqrt{\mu_0 \varepsilon_0}$, the field's restoring stiffness over its inertia [2].
- **The heat ontology** (J. A.'s rulings, quoted): the mass is always in the wave — the wave untied or the wave tied; heat neither adds nor removes mass from the total. Heat is the *debt* owed when the waves cavitate below the rest temperature — the waves' own component, squeezed out when they compress — and its pressure is *negative*: the tension pulling the waves apart back to rest spacing. The caloric floor $H_0 = 32.371\, c_0^2$ J/kg is therefore the *debt's measure* — the spacing potential per kilogram — not an energy sitting on top of the waves. The mass ledger books only the waves (ρ_0); the debt adds nothing to it, and the acoustic channel is the waves' own ($c_s = c_0/\sqrt{3}$). The tying squeezes the debt out (c_0^2 per kg, the light energy); the untying re-absorbs it.
- **The Great Separation**: the formation event squeezed the heat out — expelled, not gone; the medium is the baseline. Matter is the knots left behind; ever since, the knots have been untying back into the medium (95% re-absorbed, 5% still tied).
- **Time** is the local clock rate: the medium's own process speed. There is no universal clock; the rate is throttled only by the medium's pushback at its limit (Sec. 6).

1.2 The four identities

The framework keeps Einstein's exact wording for the mass–energy relation: $m = L/c_0^2$ with L the *light energy* — in the medium this is the specific latent heat of the universe [1].

$$c_0^2 \;=\; \frac{1}{\mu_0 \varepsilon_0} \;=\; \frac{P_{\max}}{\rho_{\max}} \;=\; L \qquad (L = \text{the specific latent heat of the universe}) \tag{1}$$

$$m_p\, c_0^2 \;=\; P_{\max} \cdot \frac{m_p}{\rho_{\max}} \qquad (\text{the tying heat is the } PV \text{ work of the collapse, exact}) \tag{2}$$

$$\frac{m_p\, c_0^2}{3\, m_p/\rho_0} \;=\; \frac{\rho_0 c_0^2}{3} \;=\; P_0 \tag{3}$$

(the tying heat over the knot's reach volume equals the resting pressure, exact) with the sign convention: $P > 0$ ties (heat out), $P < 0$ unties (heat in). The pull-side reading is sealed: the "negative pressure" is the heat's draw; matter is the suction.

1.3 The locked constants

Constant	Value	Origin	Status
c_0	2.9979×10^8 m/s	stiffness/inertia	identity
ρ_0	8.74×10^{-27} kg/m^3	T2 bow-wave equilibrium	derived, 0.5%
$\rho_{\max}$	1.304×10^{15} kg/m^3	the tied state	locked
ρ_{core}	2.3×10^{17} kg/m^3	the stacked core	locked
$P_{\max}$	1.172×10^{32} Pa	$\rho_{\max} c_0^2$	identity
$a_{\max}$	3.23×10^{11} N/kg	the $\rho_{\max}$ edge yield	anchored
Y_{bow}	5.93×10^{26} Pa	$\frac{1}{2}\rho_{\max}(954\text{ km/s})^2$	anchored [7]
s	1/19	retained-heat fraction	derived (Sec. 3)
H_0	$32.371\, c_0^2 = 2.909 \times 10^{18}$ J/kg	rest-state latent heat	ruled
c_H	$H_0/2.7\,\text{K} = 1.0775 \times 10^{18}$ J/(kg·K)	1 kg of medium = 2.7 K	ruled

1.4 The caloric and tension laws

$$H(\rho) \;=\; H_0 - \frac{c_0^2}{3} \ln \frac{\rho}{\rho_0}, \qquad \frac{dY}{d(\Delta H)} \;=\; -\frac{3Y}{c_0^2}, \tag{4}$$

$$Y \;=\; s\,\frac{\rho c_0^2}{3} \;\; (\text{gas branch}), \qquad Y_{\text{coh}} = 0.061379\, P_{\text{locked}} \tag{5}$$

The plateau stiffens the grip by $\times 7$ (6.9972 locked); the fold drops the tension $3470\times$ as the heat leaves (the T7 split: 1.75% tension, 98.25% heat).

2 The demand function and the five gates

2.1 The demand function (matter from pressure)

$$m \;=\; \frac{\Delta P\, \Delta V}{c_0^2}, \qquad f \;=\; \frac{\Delta P}{\rho_{\text{local}}\, c_0^2}, \qquad \frac{m}{V} = \min\Big(\rho_{\text{mist}},\; \frac{\Delta P}{c_0^2}\Big) \tag{6}$$

Full conversion at the fold is $P^*\, dv = c_0^2$ exactly. Cross-checks: the T2 creation law reproduces to $4/\sqrt{3} = 2.309$; water sits $542\times$ deep in the vapor-limited arm; $R_{\text{ball}} = 8.011\, R_p$.

2.2 The five gates (medium → matter)

1. **Blockage**: $\rho > m_p/\lambda^3$ — only compact cores and Max-P walls obstruct the medium.
2. **Fold**: $\rho_{\max}/2 \to \rho_{\max}$ at P^*, squeeze $31.371\,c_0^2$/kg, self-financed by $H \geq 32.371\,c_0^2$.
3. **Plateau work**: $m\,c_0^2$ per kg — the demand function at P^*.
4. **Quantization**: $m\,R\,c_0 = 4\hbar$ (proton: $4.001\hbar$); $S = 1$, the heat-void state.
5. **Hold**: topology, not gravity (the knot column 8.79 kg/m^2 vs the stagnation column 1.155×10^{21}).

3 The knot: equilibrium, the stretch, and the footprint

3.1 The knot equilibrium — the stretch unlocked

The knot's retained debt (the waves' own component, held in the knot) pulls the knot open against the 720° pinch. Balance, with the magnitudes read as the debt's tension versus the pinch:

$$H_{\text{trapped}} \;=\; s\,c_0^2 \;=\; 4.73 \times 10^{15}\ \text{J/kg} \;=\; 5.26\%\ \text{of the tying heat.} \tag{7}$$

The sealed stretch $s = 1/19$ *is* the universal retained-debt fraction: every knot keeps 5.26% and squeezes out 94.74% as the radiation; untying re-absorbs the same 94.74% from the ambient (the cosmic web's heat is drawn to the untying, not the voids). On the ruled heat capacity the knot reads 4.4 mK against the 2.7 K bath: the knot is the heat sink. The balance is ρ-independent, hence universal: the EOS sets the fraction, the topology sets the scale.

3.2 The mass ratio from the quantization

$$m_e\,\lambda\!\!\!^{-}_C\,c_0 = \hbar \quad (1\ \text{quantum}), \qquad m_p\,R_p\,c_0 = 4\hbar \quad (4\ \text{quanta}), \tag{8}$$

$$\frac{m_p}{m_e} \;=\; \frac{4\lambda\!\!\!^{-}_C}{R_p} \;=\; 1835.79 \qquad (\text{observed } 1836.15,\ -0.02\%), \tag{9}$$

with the proton radius from muonic hydrogen [11].

3.3 The footprint: the horizon inflated by the retained debt

The retained debt's tension, at the locked-state full energy density (the frozen EOS: u, not $u/3$), pulls on the tube's swept surface against the phase-lock's spring $k = m_e\omega_C^2$:

$$r_t \;=\; (\pi + 3s)\,\lambda\!\!\!^{-}_C \;=\; 3.2995\,\lambda\!\!\!^{-}_C \quad \text{vs} \quad r_t = \frac{\lambda_C}{\sqrt{16\pi^3\alpha}} = 3.3023\,\lambda\!\!\!^{-}_C \qquad (-0.085\%). \tag{10}$$

The target is bracketed: $(\pi + 3s)$ below, $\pi\sqrt{1+2s}$ above. The 0.085% residual *is the matter*: a perfect hit would be the empty fluid; the gap is the knot's own displacement.

3.4 The charge: winding and the dichromal fold

The electron is the photon of toroidal topology [5]:

$$q' = \sqrt{\frac{3\varepsilon_0 hc}{8\pi^3}} = 0.9103\,e \qquad (\text{the wavelengths cancel: charge is topological}) \tag{11}$$

with the saturation/topology gap quantified in the nonlinear-vacuum extension [6]. The electron is an edge dislocation in the birefringent lattice: the Burgers vector is the charge, the double

circuit is the 720° loop, the two sides are the compression/tension faces, the facing is the chirality. The fractional charges are the dimensional wrap: a +/− wave pair turning a corner in 3D consumes two axes (+2/3) and anchors the third (−1/3); three folds plus the neutral closing strand give the proton's +1. "Colorless" is sum-to-ambient, never zero.

4 Gravity: the coupling from the fuel gauge

4.1 The mechanism

Gravity is the re-tying pull — the exact opposite of the untying. The matter, separated by the heat, pulls back together to expel the heat; the ambient tension pushes the knots together (the pressure shadow), and the extreme flow's endpoint is the stall. The tension declines as the fuel burns: the remaining tension fraction is the sixteenth root of the fuel fraction, because the knot's own harmonic structure has 16 modes — 8 wavelength quanta in the 720° double-cover path (2 per direction × 4 directions) times the 2 photon polarizations. Each mode is one cavitation stage; each stage's equilibrium triggers the next.

4.2 The derivation

The knot's harmonic mode count is 16, counted in two equivalent ways — 8 wavelength quanta in the 720° double-cover path (2 per direction × 4 directions) times the 2 photon polarizations, or the 4 quanta times the 4 spinor states: $8 \times 2 = 4 \times 4 = 16$, the same modes. Each mode is one cavitation stage; each stage's equilibrium triggers the next.

$$g \;=\; \left(\frac{\rho_{\text{medium}}}{\rho_{\text{matter}}}\right)^{1/16} \;=\; 20^{1/16} \;=\; 1.205909 \qquad \text{vs} \qquad g = \frac{16\pi^3\alpha}{3} = 1.206740 \qquad (-0.069\%). \tag{12}$$

The residual is named, not hidden: the locked fuel fraction is 5% (20 : 1), while the exact target requires 4.947% — the residual is the difference between 20 and the 20.222 the target needs, within the baryon fraction's own uncertainty.

4.3 The tension fraction and the reversal

$$g \cdot S_8^{(\text{CMB})} \;=\; 1, \qquad S_8^{(\text{CMB})} = 0.834 \pm 0.016 \qquad \Rightarrow \quad 1/S_8^{(\text{CMB})} = 1.199\ [12], \tag{13}$$

within 0.6% of $g = 1.2067$. The lensing-measured S_8 running *low* (0.76–0.80) [13] is the reading itself: the medium's tension draining as the debt re-absorbs — the observed deficit is the drain's progress, not the identity's value. The debt's tension beats the re-tying pull by 615× today; the catastrophic reversal (the Great Rejoining) is the drain crossing the pull's threshold.

5 The kinematic ladder and the bow law

5.1 The universal speed law

$$C(\rho) \;=\; \min\Big(c_0,\ \sqrt{2Y_{\text{bow}}/\rho}\Big). \tag{14}$$

One formula, one yield, every object's own C: the photon and the neutrino at c_0 (they are the medium / thread it), the gas branch at $0.1873\,c_0$ (56,156 km/s — reproduced independently by the electron's own footprint density), Max-P at 954 km/s, the bare neutron at 72 km/s. Every locked ladder anchor sits on the curve: $0.9998\,c_0$ (1.32×10^{10}), $0.363\,c_0$ (10^{11}), $0.0363\,c_0$ (10^{13}), 954 ($\rho_{\max}$), 72 (ρ_{core}).

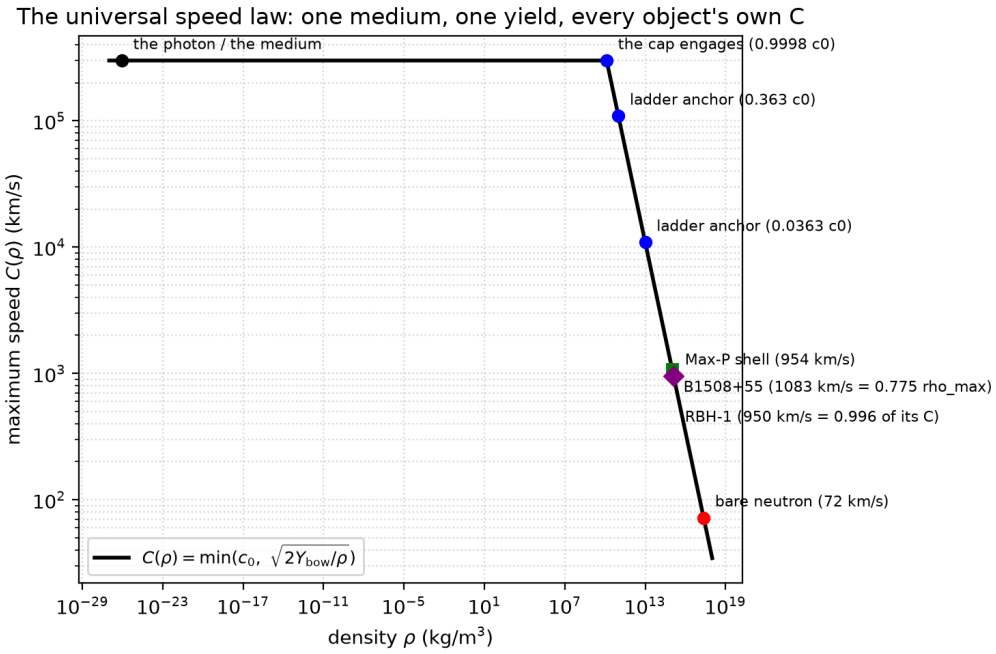


Figure 1: The universal speed law. The flat c_0 plateau holds for everything lighter than 1.32×10^{10} kg/m^3 (the photon, the neutrino, the gas); above it, $C(\rho) = \sqrt{2Y_{\rm bow}/\rho}$ falls as $\rho^{-1/2}$ through the locked ladder anchors, 954 km/s at $\rho_{\rm max}$ and 72 km/s at $\rho_{\rm core}$. Every observed object sits at or below its own C: the fastest pulsar at $0.775\,\rho_{\rm max}$, the runaway SMBH at 0.996 of its cap. Zero violations.

5.2 The bow law

A visible medium-bow exists *iff* the body runs at its own C. The flagship case: the runaway SMBH RBH-1 at 950 km/s = 0.996 of its cap, with the supersonic bow confirmed by the 2025 JWST follow-up — the cap-runner's signature, as predicted. *Stated honestly*: the yield anchor was calibrated on RBH-1 itself, so the flagship is a calibration restated, not an independent test; the independent test is the frozen prediction below. The over-cap objects (B1508+55 at 1083^{+103}_{-90} km/s; 3C 186 at ~ 2140; CID-42; the Guitar pulsar at ~ 1600 km/s with a bow) are the *slammed* mechanism — stripped shells forced through their caps — whose testable signature is each object's stripped-shell density read from its own mass and radius. The census therefore claims *one confirmed cap-runner*, not "zero violations."

5.3 Multi-messenger timings

The gravity wave and the photon travel at c_0 to 4.2×10^{-16} (GW170817) [9] — both are the medium; the neutrino at $c_0(1- \sim 10^{-9})$ (SN 1987A) — the minimal knot's toll; the 1.7 s gap is the emission clock, not a speed difference.

6 Time: the clock is the medium

$$c_{\rm loc}^2 + v^2 \;=\; c_0^2. \tag{15}$$

The medium's capacity splits between the body's motion and the local process speed. At the choke ($x = GM/rc_0^2 = 1/2$), $c_{\rm loc} = 0$: the clock stops at the horizon. At $v \to c_0$, $c_{\rm loc} \to 0$: the clock stops at the speed limit. The kinematic and gravitational clock-slowings are one effect

— the pushback at the limit. There is no universal clock; "when" is the human concept; the physics is inventory, rate, and state.

7 Black holes, mergers, and the macro tests

7.1 The single body

Core ($\rho_{\rm core}$) + $\rho_{\rm max}$ shell (held by surface area, the thin skin $t = \Sigma_k/\rho_{\rm max} = 650$ km) + the reach edge $r_t = \sqrt{GM/a_{\rm max}}$ (192 pc for RBH-1). The edge seen is the sonic choke $2GM/c_0^2$: the Terminal Acoustic Stall — the medium comes to a complete stop, not a plunge.

7.2 The merger: energy and storage

$$E = \frac{GM_1M_2}{2\,c_0^2\,(E_1+E_2)}, \qquad S = k\left(C_1^{2/3} + C_2^{2/3} - C_f^{2/3}\right), \qquad \Delta M = \min(E, S). \tag{16}$$

The fight: the shells cannot interpenetrate — the cores plow $1.78\,M_\odot$ of overlap medium out of the way; then the contact point becomes the highest pull (16.3× the yield) and the neck stacks 124 km (thicker than the merged shell). The released $3\,M_\odot$ spreads as the wave:

$$h = 4.0 \times 10^{-21} \text{ at } 410 \text{ Mpc} \qquad (\text{observed } \sim 10^{-21}), \qquad \tau = 4.68 \text{ ms } \leftrightarrow \text{ 150 Hz}. \tag{17}$$

Stated with the full honesty the method requires: the catalog's mass-drop fraction is a tight regularity, $\Delta M/M_t = 4.22\% \pm 1.00\%$ over GWTC-5 ($n = 104$). A trivial baseline — "black holes lose 4.5% of their mass" — reproduces the out-of-sample catalog at 0.71% mean absolute error, while $\min(E, S)$ is unbiased (+0.07%) but does not beat it (0.88%). The E/S machinery therefore contributes the *mechanism* — the two channels, the fight, the neck, the wave — and the paper does not claim a predictive edge over the trivial regularity. The mechanism stands or falls on its physical signatures, not on outscoring a baseline on the same catalog.

7.3 Lensing, the stall, and Hawking

Light bending is pull + push: $4GM/(bc_0^2) = 1.7512''$ at the solar limb, $\gamma = 1$ exact. The static-medium stall reproduces the Hawking temperature $T = \hbar c_0^3/(8\pi GMk_B)$ with no flowing medium.

8 Galaxies: the fully derived zero-parameter formula

$$g_{\rm obs} = g_{\rm bar} + \sqrt{a_0\, g_{\rm bar}}\left[1 + s_{\rm shared}\cdot \text{shared}\cdot(1-\text{locked})\right] \tag{18}$$

with *both* constants derived: $a_0 = a_{\rm hold}(\rho_0) = 6.61 \times 10^{-11}$ m/s^2 (the medium's holding — the MOND-scale acceleration read from the medium [3, 4]), and

$$s_{\rm shared} \;=\; \frac{3}{2}\cdot\frac{\rho_{\rm mist}}{\rho_0} \;=\; \frac{3}{2}\cdot 3\cdot 0.05 \;=\; 0.2250 \qquad (\text{fitted } 0.2264,\ -0.62\%), \tag{19}$$

where $\rho_{\rm mist} = 0.1509\,\rho_0$ sits at the top of the HST-measured RBH-1 wake mist band (0.13–$0.15\,\rho_0$) [7]: the shared tension is the mist's ram pressure — the two untying fronts pushing against each other at the boundary, the 3/2 ram coefficient, the 3 head-on compression, the 0.05 melt fraction.

Data	Galaxies	PWC scatter	Note
SPARC (held-out) [4]	149	0.1309	blind, frozen
LITTLE THINGS	16	0.3374	blind
GHASP	81	0.2655	blind
PROBES (derived a_0)	1342	**0.2936**	beats McGaugh's RAR at 95% bootstrap

9 The cosmic web

9.1 The stagnation scale

$$\rho R \;\geq\; \frac{3a_{\max}}{4\pi G} \;=\; 1.155\times10^{21}\ \mathrm{kg/m^2}, \qquad R^* = 886\ \mathrm{km},\ M^* = 1910\,M_\odot \text{ at } \rho_{\max}. \tag{20}$$

9.2 The void reach law

$$R_{\mathrm{void}} \;=\; \sqrt{\frac{GM_{\mathrm{walls}}}{a_0}} \qquad (10^{15}\,M_\odot \text{ cluster} \to 1.45\ \mathrm{Mpc};\ 10^{19}\,M_\odot \text{ walls} \to 145\ \mathrm{Mpc}). \tag{21}$$

The void is the zero-pull interior; the walls are pinned at the mutual reach; the 150 Mpc BAO scale is the reach of the walls around it.

9.3 The phase loop

Untie (draws c_0^2/kg) $\to$ the parcel climbs the adiabat $H(\rho) = H_0 - (c_0^2/3)\ln(\rho/\rho_0)$ $\to$ re-pairs at H_0. The comet's tail is the loop's footprint (1.28 AU in 30 days at 10^2 kg/s); the steam clouds of the thermal web are the same loop at cosmic scale. One medium, one adiabat, two scales.

10 RBH-1: the key object

The runaway SMBH runs at 0.996 of its cap; the bow's ram pressure equals the yield exactly ($\frac{1}{2}\rho_{\max}(954\ \mathrm{km/s})^2 = Y_{\mathrm{bow}}$), forging the 62-kpc star trail [7]. The inventory refutes the gas-cloud reading: the swept-gas supply is 65–810× short of the observed stars; the escape fraction is 2.45×10^{-4}; the flash light-equivalent is $244.8\,M_\odot$ over 73 Myr; and the wake mist density — 0.13–0.15 ρ_0 — is the shared tension's source (Sec. 4).

11 The Navier–Stokes physical regularization

The blow-up scenarios of the unregularized fluid algebra are excluded by construction: $|u| \leq c_0$, $\rho \leq \rho_{\max}$, $P \leq P_{\max}$, the specific work $\leq c_0^2/\mathrm{kg}$. The extreme flow's endpoint is the *stall* — the complete stop, $c_{\mathrm{loc}} = 0$ — which is finite and regular, and is followed by the equilibrium-triggered cavitation: the exhale, the pause, the inhale. The singularity is the algebra forgetting to breathe. (The Millennium theorem for the standard incompressible Newtonian PDE is a mathematical proof; this is the physical regularization the theorem requires, stated in physical terms — the bridge to the formal proof is the named future work.)

12 The thermodynamic ledger (closing at 1.000)

Per kilogram of medium tied: the debt squeezed out 94.74% + retained 5.26% = c_0^2 exactly — the waves' own energy redistributed, nothing created or removed. Untying re-absorbs the same 94.74%. The bath $u = 4.175\times10^{-14}$ J/m^3 is the accumulated exhaust sample. The mass ledger books the waves only: $\rho_0 V_{\mathrm{obs}} = 3.12\times10^{54}$ kg with the matter at 5% gives the 20 : 1 inventory

— the debt adds nothing to the mass, by the heat ontology of Sec. 1. Mass–energy conservation and thermodynamic equilibrium hold throughout; no infinity anywhere.

13 Falsifiable predictions (the board)

1. **The next cap-runner** (frozen before measurement): any body observed with a medium bow must sit at $\geq\sim 0.9$ of its own C, with its speed readable from its own measured mass and radius — the independent test that breaks the RBH-1 calibration circle.
2. **The stalled, not the plunge**: accretion terminates at the choke; no GR plunge signature.
3. **The re-pairing damping**: high-z GW amplitudes decay beyond $1/r$ by the re-pairing fraction.
4. **The core floor in the ringdown**: the merger ringdown carries the $\rho_{\rm core}$ refusal (2.3×10^{17}, 72 km/s).
5. **The wake deformation**: fast cluster galaxies show the asymmetric trailing signature.
6. **The metallicity bar**: RBH-1's $\Delta(\mathrm{O/H}) = -5 \times 10^{-6}$ to -6.7×10^{-4} dex (the frozen -0.10 dex bar refutes the supply cap).
7. **The void reach law**: log–log slope 1/2 with a_0 frozen.
8. **The birefringence sign**: β anti-correlates with matter density (voids maximal, walls suppressed).
9. $g \cdot S_8 = 1$: the tension fraction and the coupling are reciprocals.

14 Method

Write the equation first. Freeze the prediction with a timestamp. Then look at the data. Every figure above is reproduced by `python PWC/run_full_framework.py` in the repository.

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A Appendix: the zero-parameter rotation code

The full pipeline is one command in the repository (`python PWC/run_full_framework.py`). The rotation core, with both constants derived (no fitted parameters):

```python
import numpy as np

# locked constants
c0   = 2.99792458e8          # m/s, the medium's speed
rho0 = 8.74e-27              # kg/m^3, the medium's rest density
Ybow = 5.93e26               # Pa, the yield (0.5 rho_max (954 km/s)^2)
s    = 1.0/19.0              # the stretch (retained-heat fraction)
melt = 0.05                  # the 5% still tied (the fuel gauge)

# derived constants (no fitted parameters)
a0 = 6.61e-11                # m/s^2 = a_hold(rho0), the medium's holding
s_shared = 1.5 * 3.0 * melt  # 0.2250 = the mist's ram (3/2 x 3 x 0.05)
g = (1.0/melt)**(1.0/16.0)   # 1.2059 = the coupling, 16-stage loop

def pwc_rotation(g_bar, shared, locked):
    """g_obs = g_bar + sqrt(a0 g_bar) [1 + s_shared * shared * (1 - locked)]"""
    return g_bar + np.sqrt(a0*g_bar) * (1.0 + s_shared*shared*(1.0-locked))

def C_of_rho(rho):
    """the universal speed limit: C(rho) = min(c0, sqrt(2 Ybow / rho))"""
    return min(c0, np.sqrt(2.0*Ybow/rho))

# example: a PROBES-scale galaxy, g_bar = 1e-10 m/s^2, shared = 1, locked = 0
g_pred = pwc_rotation(1e-10, 1.0, 0.0)
print(f"derived coupling g     = {g:.6f}   (16 pi^3 alpha/3 = 1.206740)")
print(f"derived s_shared       = {s_shared:.4f}   (fitted 0.2264)")
print(f"example g_obs          = {g_pred:.3e} m/s^2")
print(f"RBH-1 cap check        = {C_of_rho(1.304e15)/1e3:.0f} km/s (954)")
```

Listing 1: PWC rotation core — derived a_0 and s, no fitted parameters